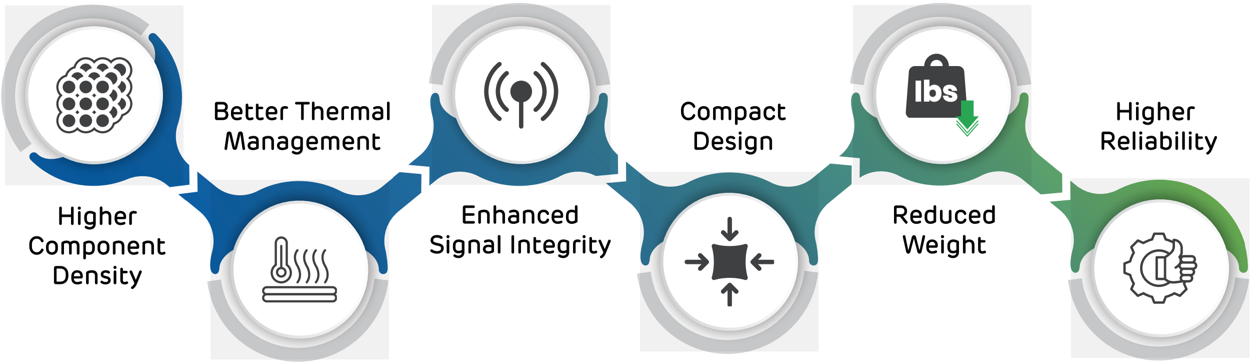
MULTILAYER PCBs

From simple builds to complex, high-layer-count constructions built to demanding specifications Excello Circuits delivers consistency & reliability at every stage. With advanced manufacturing capabilities & disciplined process control, we support accelerated production without compromising quality.

KEY BENEFITS



DESIGN CAPABILITIES

01.

LAYER STACK-UP FLEXIBILITY

Choose from our standard stack-ups, or we can accommodate custom stack-ups you provide.

02.

HIGH-DENSITY ROUTING

Our capabilities support ultra-fine traces & spacing for complex, compact circuits.

03.

IMPEDANCE CONTROL

Strict impedance tolerances are maintained to ensure signal integrity in high-speed designs.

04.

ADVANCED VIA OPTIONS

Blind, buried, & via -in -pad to reduce form factors & improve performance.

05.

WIDE SELECTION OF MATERIALS

From FR4 to Polyimide, Rogers & many more - suiting diverse application requirements.

TECHNICAL SPECIFICATION

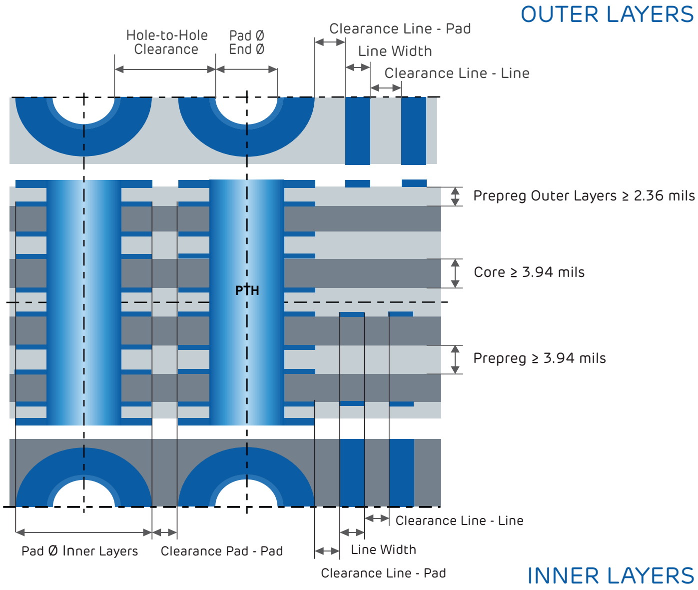
TECHNICAL SPECIFICATION	DETAILS
Minimum Trace Width / Spacing	2.75 mils (After Etch Compensation)
Minimum Microvia Size	3.0 mils
Layer Count	up to 32 layers
Aspect Ratio for Microvias	1:1
Max. Aspect Ratio for Through-Hole Vias	15:1
Copper Thickness	up to 20 oz.
Dielectric Thickness	down to 2.0 mils
Impedance Control Tolerance	+/-10%
Finished Board Thickness	0.006" to 0.250"

COPPER STRUCTURES, SPACINGS

EXTERNAL LAYER CAPABILITIES		
Copper Finished Thickness	Minimum Line Width (mil)	Minimum Space (mil) (After Etch Compensation)
0.5 oz	2.75	2.5
1.0 oz	3.5	3.0
2.0 oz	4.5	4.0
3.0 oz	6.5	5.0
4.0 oz	7.5	6.0
5.0 oz	9.0	10.0
6.0 oz	11.0	20.0
7.0 oz	13.0	22.0
8.0 oz	15.0	24.0

INTERNAL LAYER CAPABILITIES		
Starting Copper Weight	Minimum Line Width (mil)	Minimum Space (mil) (After Etch Compensation)
0.5 oz	2.75	2.5
1.0 oz	3.5	3.0
2.0 oz	4.5	4.0
3.0 oz	6.5	5.0
4.0 oz	7.5	6.0

The copper spacing, often referred to as conductor spacing, is a critical parameter in the production of copper structures. It impacts various design features, including trace-to-trace, trace-to-shape, shape-to-trace, shape-to-shape, trace-to-all pin pads, trace-to-all via pads, & trace-to-all non-signal geometries, ensuring precision & reliability in manufacturing.



DRILLS, PADS, ANNULAR RINGS, CLEARANCES

PLATED THROUGH HOLES					
Pad Size	Aspect Ratio	Drill Tool Diameter	Finished Hole Diameter	Tolerance (Standard)	Copper Clearance Plane on Inner Layer Without Pad (mil)
0.018"	15:1	0.008"	0.004”	+/-3.0 mils	Edge of hole +8.0 per side min.
0.024"	15:1	0.012"	0.008”	+/-3.0 mils	Edge of hole +8.0 per side min.
0.050"	15:1	0.020"	0.016”	+/-3.0 mils	Edge of hole +8.0 per side min.

DRILL DIAMETER

The PCB design defines the via hole size & the via pad size, which determine the finished hole diameter. The drill tool diameter is selected larger than the finished hole diameter to account for material removal during drilling & copper plating. Proper pad size selection ensures sufficient copper clearance & structural reliability.

SPACING BETWEEN HOLES

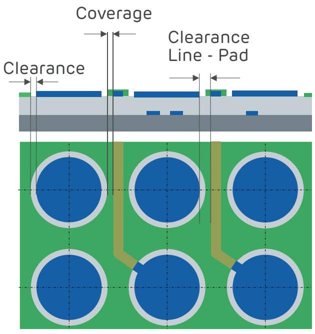
Minimum spacing requirements based on final diameters:

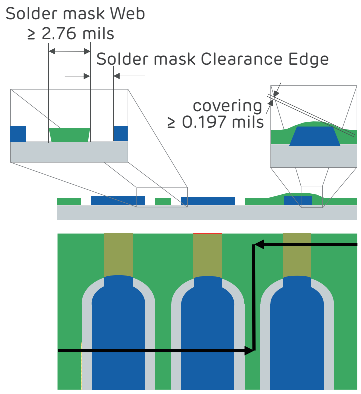
Hole-to-hole clearance (Same potential):	Edge of hole +12.0 mils
Hole-to-hole clearance (Different potential):	Edge of hole +12.0 mils
NPTH-to-NPTH clearance:	Edge of hole +10.0 mils

SOLDER MASK DESIGN SPECIFICATIONS

Manufacturing without solder mask clearances requires additional effort & is not recommended, as it may compromise overall quality.

PARAMETER	SPECIFICATION
Min. clearance	1.0 mil / side (pad size +2.0 mils)
Coverage	1.0 mil
Solder mask web	4.0 mils preferred, 3.0 mils minimum (Green)
Via-opening	10.0 mils





COPPER LAYER THICKNESS IN PTHS, BLIND & BURIED VIAS

Copper thickness in plated through holes & vias complies with IPC-6012 standards :	Tolerance: +/-3.0 mils
Copper clearance on inner layers without pads :	Edge of hole to copper feature 10.0 mils in design

SPACING COPPER TO CONTOUR

Spacing requirements for copper to PCB contour:

Routed board edge:	6.0 mils
V-scored board edge:	≥4.0 mils for board thickness ≥0.126"

LEGEND DESIGN SPECIFICATIONS

PARAMETER	SPECIFICATION
Printed legend minimum stroke / width	4.5 mils
LPI legend capability	Yes
LPI legend minimum stroke / width	4.0 mils
Screened / LPI legend colors	White, Black, Yellow, Red, Blue
Serialization / Unique Serialization	Yes